

3D ANN CLOUD COMPUTING GRAPHICS PROCESSOR

- Operational, kinetic and implementation demonstrator
- Virtual 1000MB x 1000MB x 1000MB logical 3D field
- Sparse allocation: only active chunks occupy physical memory
- Recursive encode -> ANN compute -> graphics -> decode -> verify
- Cloud placement and optimisation close the loop

1. SYSTEM ARCHITECTURE

- Input is chunked, hashed and mapped to stable 3D coordinates.
- The same identity follows data through storage, ANN, memory and graphics.
- Runtime modules: ChunkStore, SpatialIndex, TensorRuntime, ANNCore, Graphics, Scheduler, Telemetry, Codec, Verification.
- Physical allocation is bounded by active-set size, not virtual cube size.

2. 3D VIRTUAL ADDRESS SPACE

- $x = i \bmod N_x$
- $y = \text{floor}(i/N_x) \bmod N_y$
- $z = \text{floor}(i/(N_x*N_y))$
- $i = x + N_x*(y + N_y*z)$
- A_t = active coordinates only; empty logical regions allocate nothing.

3. END-TO-END KINETIC CYCLE

- $M_t = (X, Z, H, G, \Omega, \Theta, R)$
- Acquire -> Map3D -> Encode -> Latent -> ANN -> Graphics -> Decode -> Error -> Learn -> Repeat
- Error updates parameters, memory, routing and scheduling.
- The state evolves as one recurrent system rather than isolated stages.

4. RECURSIVE INWARD ENCODER

- $X^{(l+1)} = \sigma(W_l * X^{(l)} + b_l)$
- $Z_{(k+1)} = E_k(Z_k, \Omega)$
- Stop when $\|Z_{(k+1)} - Z_k\| < \epsilon$ or budget expires.
- Inward means spatial contraction plus semantic abstraction, not physical motion.

5. 3D ANN PROCESSING CORE

- Local sites combine hidden state, neighbours, latent context and memory.
- $H_{(t+1)}[p] = \sigma(W_0 H_t[p] + \sum_d W_d H_t[p+d] + U Z_t)$
- Global attention connects distant regions: $A = \text{softmax}(QK^T/\sqrt{d} + B_{3D})V$.
- Conditional routing activates only required experts and kernels.

6. MEMORY AND PERMEATION

- $\Omega_t(x,y,z)$ stores persistent or recurrent local state.
- $\Omega_{t+1} = \lambda \Omega_t + W_h H_t + W_e E_t$
- Coordinate identity and provenance permeate encoder, ANN, graphics and cloud execution.
- Memory promotion is selective and bounded.

7. GRAPHICS PROCESSING PATH

- Classical: ANN state -> geometry/rays -> shading -> frame.
- Neural: $\text{Frame} = R_Theta(Z, H, \text{Camera}, \text{Light})$.
- Rendered observations can feed back into ANN state.
- Use cases include simulation, digital twins, scene reconstruction and interactive inference.

8. CLOUD PARTITIONING

- $V = \bigcup_p V_p$; each worker owns active bricks.
- π^* : regions $\rightarrow \{\text{CPU, GPU, NPU, EDGE, CLOUD}\}$
- Placement minimises latency, memory pressure, energy, monetary cost and network traffic.
- Logical addresses remain stable while physical placement migrates.

9. AUTO-OPTIMISATION

- $\Theta_{t+1} = \Theta_t - \eta * \text{grad } L_t$
- $R_{t+1} = O(R_t, \text{telemetry}_t)$
- Tune precision, sparsity, batch size, kernels, cache policy, routing and placement.
- Promote topology changes only after measured, reversible benchmark wins.

10. OUTWARD DECODER AND VERIFICATION

- $X_{\text{hat}} = D_{\text{Theta}}(Z, H)$
- Lossless chunks require $\text{Hash}(X_i) == \text{Hash}(X_{\text{hat}_i})$.
- Learned reconstructions require task-specific error and uncertainty thresholds.
- Universal arbitrary 1GB-to-tiny lossless compression is impossible; entropy still applies.

11. WORKED 1 GB PAYLOAD

- Split about 1 GB into about 1000 x 1 MB chunks.
- Assign each chunk coordinate, hash, codec and manifest metadata.
- Stream only required chunks into active tensor tiles.
- Process -> decode -> verify each chunk -> emit output and provenance.

12. PERFORMANCE MODEL

- $T_{total} \approx T_{io} + T_{encode} + T_{ann} + T_{graphics} + T_{decode} + T_{sync}$
- Memory bandwidth and cache locality often dominate.
- Sparse kernels win only when sparsity structure offsets indexing overhead.
- Network collectives, attention and routing can dominate distributed execution.

13. IMPLEMENTATION BLUEPRINT

- Minimum loop: ingest -> chunk/hash -> sparse map -> encode -> ANN -> render -> decode -> verify.
- Start working, then robust, portable, optimised, advanced.
- Benchmark against dense CPU/GPU baselines before adding speculative mechanisms.
- Keep storage, compute, graphics and verification separately testable.

14. UNIFIED OPERATIONAL FORMULATION

- $M_{(t+1)} = \text{Pi_Lambda}(F_Theta(M_t, \text{telemetry}_t, \text{error}_t))$
- Acquire -> Spatialise -> Encode inward -> Compute -> Render -> Decode outward -> Verify -> Learn -> Reschedule -> Repeat.
- Reality check: virtual capacity != physically allocated memory.
- Actual scale is bounded by active work, accelerator memory, network bandwidth and data entropy.